

MATERIALS ENGINEERING

For undergraduate curriculum in materials engineering leading to the degree Bachelor of Science. The materials engineering program is accredited by the Engineering Accreditation Commission of ABET, <https://www.abet.org> (<https://www.abet.org/>), under the commission's General Criteria and Program Criteria for Materials (1), Metallurgical (2), Ceramics (3) and Similarly Named Engineering Programs. Materials engineering is a broadly-based discipline focusing on processing, structure, properties, and performance of all types of advanced materials aimed at virtually every possible application.

Because of its interdisciplinary nature, career opportunities for materials engineers bridge all sectors of industry, technology, applied R&D, and fundamental research, including: advanced manufacturing and processes, materials for energy and power, biomaterials and biomedical technology, functional materials and devices, nanomaterials and coatings, structures, durable goods, infrastructure, and vehicle technologies (automotive, aerospace).

Student Learning Outcomes

Graduates of the materials engineering curriculum should have at the time of graduation:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences.
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Program Educational Objectives

Within a few years of completing the materials engineering degree program, our graduates are expected to:

- *Work*-competently engage in professional employment, applying knowledge and skills related to the structure, properties, processing, and performance of materials.
- *Learn*-invest in broad life-long learning activities, such as advanced academic or professional degree programs, practical training, professional development, and leadership training.
- *Serve*-actively contribute to professional communities in their chosen career paths, including professional societies, advisory boards, standards and regulatory boards, and internal company organizations.

Graduates in materials engineering are able to apply scientific and engineering principles to the performance-based design of materials, processes, devices, and systems. With a robust program integrating classroom learning with a comprehensive hands-on laboratory experience, students learn advanced methods for materials processing, microstructure control, microchemical analysis, structural characterization, property testing, and performance optimization. Graduates have the opportunity to gain experience in materials engineering practice through cooperative work experience or internships in industry, national laboratories, or another funded research. Graduates can develop a global perspective of engineering through various study abroad opportunities.

A degree in materials engineering relies on a strong foundation of math, chemistry and physics. The core materials courses include fundamentals of materials, kinetics and thermodynamics, mechanical properties, computational methods, design, and professional practice experience. Students tailor their programs to their goals and interests through the selection of elective courses in various areas, including ceramics, metals, polymers, and biomaterials. The breadth and depth of the program provide excellent preparation for immediate entry into industry, further study in graduate school, as well as many other unique career paths (i.e. law, military service, art conservation, museum conservatorship, etc.).

Curriculum in Materials Engineering

Administered by the Department of Materials Science and Engineering.

Leading to the degree Bachelor of Science.

Total credits required: 128 cr. Any transfer credit courses applied to the degree program require a grade of C or better (but will not be calculated into the ISU cumulative GPA, Basic Program GPA or Core GPA). See also Basic Program and Special Programs.

International Perspectives: 3 cr.¹

U.S. Cultures and Communities: 3 cr.¹

Communication Proficiency/Library requirement:

ENGL 1500	Critical Thinking and Communication (Must have a C or better in this course)	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition (Must have a C or better in this course)	3
LIB 1600	Introduction to College Level Research	1
Complete one of the following courses (Must earn a grade of C or better)		3
ENGL 3140	Technical Communication	
ENGL 3090	Proposal and Report Writing	
ENGL 3020	Business Communication	

Advanced Communication Courses: 6 cr.

ENGL 2500	Written, Oral, Visual, and Electronic Composition (Must have a C or better in this course)	3
Complete one of the following courses (Must earn a grade of C or better)		3
ENGL 3140	Technical Communication	
ENGL 3090	Proposal and Report Writing	
ENGL 3020	Business Communication	

General Education Electives: 12 cr.

Complete 12 cr. from approved list with a minimum of 3 cr. from 2000+ level courses and maximum of 9 cr. from the same designator².

Basic Program: 24 cr.³

A minimum GPA of 2.00 is required for this set of courses. (Please note that transfer course grades will not be calculated into the Basic Program GPA.) See Requirement for Entry into Professional Program in College of Engineering Overview section.

CHEM 1770	General Chemistry I	4
or CHEM 1670 General Chemistry for Engineering Students		
ENGL 1500	Critical Thinking and Communication (Must have a C or better in this course)	3
ENGR 1010	Engineering Orientation	R
ENGR 1600	Engineering Problems with Computer Applications Laboratory ³	3
LIB 1600	Introduction to College Level Research	1
MATH 1650	Calculus I	4
MATH 1660	Calculus II	4
PHYS 2310	Introduction to Classical Physics I	4

PHYS 2310L	Introduction to Classical Physics I Laboratory	1
Total Credits		24

Math and Physical Science: 18 cr.

CHEM 1770L	Laboratory in General Chemistry I	1
CHEM 1780	General Chemistry II	3
CHEM 1780L	Laboratory in College Chemistry II	1
MATH 2650	Calculus III	4
MATH 2670	Elementary Differential Equations and Laplace Transforms	4
PHYS 2320	Introduction to Classical Physics II	4
PHYS 2320L	Introduction to Classical Physics II Laboratory	1
Total Credits		18

Materials Engineering Core: 50 cr.

A minimum average GPA of 2.00 is required for this set of courses.

(Please note that transfer course grades will not be calculated into the Core GPA.)

MATE 1700	Fundamentals of Computation for Materials Engineering	3
MATE 2140	Structural Characterization of Materials	3
MATE 2150	Fundamentals of Materials Engineering I	3
MATE 2150L	Fundamentals of Materials Engineering I - Lab	1
MATE 2160	Fundamentals of Materials Engineering II	3
MATE 2160L	Fundamentals of Materials Engineering II - Lab	1
MATE 3110	Thermodynamics in Materials Engineering	3
MATE 3140	Kinetics of Phase Transformations in Materials	3
MATE 3170	Introduction to Electronic Properties of Materials	3
MATE 3190	Mechanics of Structures and Materials	3
MATE 4130	Materials Design and Professional Practice I	3
MATE 4140	Materials Design and Professional Practice II	3
MATE 4180	Mechanical Behavior of Materials	3
MATE 4700	Materials Selection for Engineering Design	3
MATE Core Electives ²		12
Total Credits		50

MATE Core ELECTIVES:

Students must complete 12 cr. from the course options listed below.

MATE 3210	Introduction to Ceramic Science
MATE 3220	Introduction to Ceramic Processing
MATE 3410	Metals Processing
MATE 3420	Structure/Property Relations in Nonferrous Metals
MATE 3500	Polymers and Polymer Engineering.
MATE 3510	Introduction to Polymeric Materials
MATE 4250	Glass Science and Engineering

MATE 4330	Advanced Ceramics and Electronic Materials	
MATE 4430	Physical Metallurgy of Ferrous Alloys	
MATE 4440	Corrosion and Failure Analysis	
MATE 4530	Physical and Mechanical Properties of Polymers	
MATE 4540	Polymer Composites and Processing	

Other Courses: 18 cr.

STAT 3005	Engineering Statistics	3
MATE Electives ²		6
Technical Electives ²		9
Total Credits		18

Seminar/Co-op/Internships

MATE 3010	Materials Engineering Professional Planning	R
Co-op and internships are optional		

1. These university requirements will add to the minimum credits of the program unless the university-approved courses are also approved by the department to meet other course requirements within the degree program. U.S. Cultures and Communities and International Perspectives courses may not be taken Pass/Not Pass.
2. Choose from department approved list.
3. See Basic Program for Professional Engineering Curricula for accepted substitutions for curriculum designated courses in the Basic Program.

Note: A MATE student may take up to 6 credit hours from General Education electives on a P/NP basis, except for courses used to meet the U.S. Cultures and Communities and International Perspectives requirements. S/F courses (different from P/NP) will be considered for these requirements on a course-by-course basis.

See also: A 4-year plan of study grid showing course template by semester.

Materials Engineering, B.S.

First Year

Fall	Credits Spring	Credits
MATH 1650	4 MATH 1660	4
CHEM 1770	4 CHEM 1780	3
CHEM 1770L	1 CHEM 1780L	1
ENGL 1500	3 MATE 1700	3
ENGR 1010	R Gen Ed Elective	3
ENGR 1600	3 U.S. Cultures and Communities	3
LIB 1600	1	
	16	17

Second Year

Fall	Credits Spring	Credits
MATH 2650	4 MATE 3010	R
MATE 2150	3 MATH 2670	4
MATE 2150L	1 MATE 2140	3
PHYS 2310	4 MATE 2160	3
PHYS 2310L	1 MATE 2160L	1
ENGL 2500	3 PHYS 2320	4
	PHYS 2320L	1
	16	16

Third Year

Fall	Credits Spring	Credits
MATE 3110	3 MATE 3140	3
MATE 3170	3 MATE 3190	3
MATE Core Elective	3 MATE 4700	3
MATE Elective	3 MATE Core Elective	3
STAT 3005	3 International Perspectives	3
	15	15

Fourth Year

Fall	Credits Spring	Credits
MATE 4130	3 MATE 4140	3
MATE 4180	3 MATE Core Elective	3
MATE Core Elective	3 MATE Elective	3
Technical Elective	3 Technical Elective	3
Technical Writing	3 Technical Elective	3
	Gen Ed Elective	3
	15	18

The MSE Department offers concurrent enrollment program opportunities for students wishing to enroll in either the M.S. degree program or the M.Eng. degree program during the final stages of their B.S. program completion. Minimum admission requirements for these concurrent programs are the same as those for the regular M.S. and M.Eng. admission. Concurrently enrolled students are considered graduate students for official enrollment and tuition/fee purposes. MSE students concurrently enrolled as B.S./M.S. students are eligible for Graduate Research Assistantships (GRA), but these require Major Professor sponsorship and are not guaranteed with admission to the program. Assistantships are not available for B.S./M.Eng. students.

Undergraduate students should visit with both their academic advisor and the MSE Director of Graduate Education to discuss interest in a concurrent program, to learn more about blending the undergraduate and graduate curricula, and to obtain guidance related to the application

process. For additional details about the MSE concurrent enrollment program, please visit: <https://www.mse.iastate.edu/concurrent/>.

The MSE Department also offers a B.S. MATE and MBA concurrent enrollment program in collaboration with the Ivy College of Business. For more information about this program, please visit: <https://www.ivybusiness.iastate.edu/full-time-concurrent-mba/>.

Pursuing next-generation technologies and robust solutions in areas such as energy, transportation, infrastructure, cyber-security, manufacturing, wearables, agriculture, nanomaterials, and medicine, graduate students work with our interdisciplinary faculty to address some of the world's most critical challenges and to explore exciting new opportunities for innovation.

- **MSE Graduate Students** – Our students come from all over the world and from a wide variety of technical fields, including all engineering disciplines as well as physics, chemistry, biology, computer science, and mathematics. Career paths vary widely, and many go on to become leaders and innovators in academics, research, industry, medicine, business, and public service.
- **MSE Advanced Degree Options** – The Department of Materials Science and Engineering offers three advanced degree options, including Ph.D, M.S. and M.Eng. degrees. We also offer two concurrent enrollment options (BS/MS and BS/MEng), where qualified students can begin to pursue an advanced degree while working on the final stages of the MSE undergraduate program.
- **MSE Research Assistantships** – Nearly all of our graduate students enrolled in a thesis-based program (M.S. or Ph.D.) complete their graduate studies under some form of Research Assistantship, helping to offset all or part of their tuition and living expenses while pursuing an advanced degree.

To learn more about MSE Graduate Programs, please visit: <https://www.mse.iastate.edu/graduate-programs/>.